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DEEP LEARNING-BASED POTHOLE DETECTION AND ROAD DAMAGE ANALYSIS SYSTEM

Computer Science

Mrs. Esther Priya. R Assistant Professor, Jansons Institute of Technology.
Busireddy Lakshmi Bhavani UG Student, Jansons Institute of Technology.
Chinni Mahalakshmi UG Student, Jansons Institute of Technology.
Sasi Dharan R UG Student, Jansons Institute of Technology.
Shyaamal S UG Student, Jansons Institute of Technology.

ABSTRACT

Road surface damage such as potholes poses significant safety risks, increases vehicle maintenance costs, and disrupts transportation efficiency. Conventional inspection methods are manual, time-consuming, and unsuitable for large-scale monitoring. This paper presents an AI-based road quality monitoring and predictive maintenance system that automates pothole detection, severity classification, and damage assessment using deep learning techniques. The proposed system integrates YOLO for real-time pothole detection and a Convolutional Neural Network (CNN) for severity classification into categories such as Moderate, Poor, and Severe. Grad-CAM is utilized to provide visual explainability, enabling better understanding of model decisions. Furthermore, a Flask-based web application is developed to facilitate image uploading, visualization of results, and report generation. The system also incorporates GPS-based location tagging and structured data logging for efficient infrastructure management and decision-making. Experimental results demonstrate high detection accuracy, reliable severity estimation, and improved interpretability.

KEYWORDS

Pothole Detection, YOLO, CNN, Grad-CAM, Road Monitoring, Deep Learning

INTRODUCTION

Road infrastructure plays a crucial role in transportation systems, and its maintenance is essential for safety and efficiency. Road surface damages such as potholes lead to accidents, vehicle damage, and increased maintenance costs. Traditional inspection methods are manual, time-consuming, and unsuitable for large-scale monitoring.

With advancements in Artificial Intelligence and Computer Vision, automated road damage detection systems have gained significant attention. Deep learning-based object detection models, particularly YOLO, enable efficient real-time pothole detection [1][2][5]. However, most existing approaches focus only on detection and lack severity analysis and model interpretability [3][7][8].

This paper proposes an AI-based road surface monitoring system that integrates YOLO for pothole detection, Convolutional Neural Networks (CNN) for severity classification, and Grad-CAM for visual explainability [4], improving model interpretability. A Flask-based web application is also developed for visualization, reporting, and analysis. Additionally, GPS-based location tracking and data logging are incorporated to support smart infrastructure management[9][10].

In recent years, smart city initiatives have increased the demand for automated infrastructure monitoring systems. Efficient road maintenance requires continuous monitoring and timely identification of damages. Traditional manual inspection methods are not only labor-intensive but also prone to human error. Therefore, intelligent systems capable of real-time detection and analysis are essential for improving road safety and reducing maintenance costs.

In addition to detection accuracy, modern road monitoring systems require interpretability and scalability for real-world deployment. Explainable AI techniques help in understanding model decisions, which is critical for infrastructure applications. Moreover, integrating detection systems with web-based platforms enhances usability by enabling real-time monitoring, reporting, and data-driven decision-making for road maintenance authorities.

METHODOLOGY

The proposed system follows a structured pipeline consisting of image acquisition, preprocessing, detection, classification, and visualization. Initially, road images are collected from various sources and resized to a uniform dimension for efficient processing. Image normalization is applied to improve model performance.

The YOLO model is employed for real-time pothole detection [1][5][6], where bounding boxes are generated around damaged regions. These detected regions are then extracted and passed to a Convolutional Neural Network (CNN) for severity classification. The CNN model learns spatial features and categorizes potholes into different severity levels such as Moderate, Poor, and Severe[3][7].

To enhance model interpretability, Grad-CAM is used to generate heatmaps that highlight important regions influencing the model's predictions[4][8]. Furthermore, a Flask-based web application is developed to provide an interactive interface for uploading images, viewing detection results, and generating reports. GPS-based location tracking is also integrated to associate detected damages with geographic coordinates, enabling efficient infrastructure management.

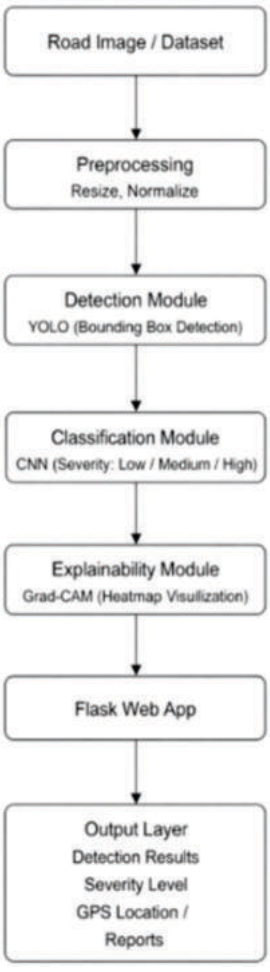


Figure 1: Pothole Detection using YOLO Model

System Architecture

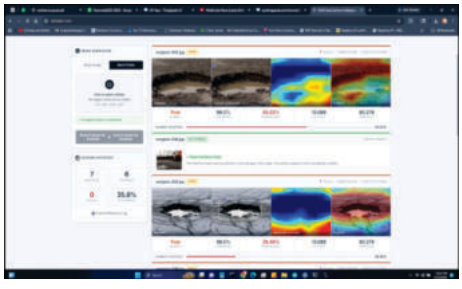
The overall architecture of the proposed system is designed to ensure efficient processing, accurate detection, and user-friendly interaction. The system consists of four main layers: data acquisition, processing, analysis, and visualization [6][9].

In the data acquisition layer, road images are collected from various sources such as mobile cameras, datasets, and real-time captures. These images are then passed to the preprocessing module, where resizing, normalization, and noise reduction are applied to enhance image quality.



The processing layer utilizes the YOLO model for detecting potholes by generating bounding boxes around damaged regions. These detected regions are further processed by a Convolutional Neural Network (CNN), which classifies the severity of the potholes into different categories.

In the analysis layer, Grad-CAM is applied to visualize important regions contributing to the model's predictions, improving interpretability. Finally, the visualization layer is implemented using a Flask-based web application, which allows users to upload images, view detection results, and analyze reports efficiently.



RESULTS AND DISCUSSION

The performance of the proposed system was evaluated using a dataset containing diverse road images with varying lighting and environmental conditions. The YOLO model demonstrated strong detection capabilities with high accuracy in identifying potholes in real-time[2][5][7]. The CNN model effectively classified pothole severity into predefined categories, ensuring reliable assessment of road conditions.

The system demonstrated robustness under different environmental conditions, including variations in lighting, shadows, and road textures. The detection model maintained consistent performance across diverse scenarios, indicating its generalization capability. The classification model also showed reliable performance in distinguishing severity levels, which is essential for prioritizing maintenance activities.

The integration of Grad-CAM provided visual explanations of the model's predictions by highlighting the most relevant regions in the input images [4][8]. This improved the transparency and trustworthiness of the system. The proposed approach showed consistent performance across different scenarios, making it suitable for real-world applications. Additionally, the Flask-based interface enabled easy visualization and reporting, enhancing user interaction and accessibility.

Advantages of Proposed System

The proposed system offers several advantages over traditional road inspection methods. Firstly, it enables automated detection of potholes, reducing the dependency on manual inspection and minimizing human error. Secondly, the use of deep learning models ensures high accuracy and efficiency in identifying road damages.

The integration of severity classification provides additional insights, allowing authorities to prioritize maintenance activities based on damage levels. Furthermore, the use of Grad-CAM enhances model transparency by visually explaining predictions, which is crucial for real-world deployment.

Another key advantage is the implementation of a Flask-based web interface, which improves accessibility and usability. Users can easily upload images, view results, and generate reports without requiring technical expertise. Additionally, GPS integration supports location-based analysis, making the system highly suitable for smart city applications.

Limitations and Future Work

Despite its effectiveness, the proposed system has certain limitations. The performance of the model may be affected by extreme environmental conditions such as poor lighting, heavy shadows, or occlusions. Additionally, the accuracy of detection depends on the quality and diversity of the training dataset.

Future work can focus on improving the robustness of the model by incorporating larger and more diverse datasets. Real-time video processing can be integrated to enhance continuous monitoring capabilities. Furthermore, deploying the system on cloud platforms can enable large-scale implementation and centralized data management.

The system can also be extended by integrating additional features such as crack detection, road surface classification, and predictive maintenance using historical data. These improvements can further enhance the effectiveness of intelligent road monitoring systems.

Applications

The proposed system can be applied in various real-world scenarios, particularly in smart city infrastructure and transportation management. It can assist government authorities in monitoring road conditions and planning maintenance activities efficiently.

The system can also be used by autonomous vehicle systems to detect road hazards in real-time, enhancing driving safety. Additionally, it can be integrated into mobile applications for crowdsourced road monitoring, where users can capture and upload road images for analysis.

The ability to provide severity classification and location-based data makes the system valuable for decision-making and resource allocation in large-scale road networks.

CONCLUSIONS

This paper presents an AI-based road surface monitoring system for automated pothole detection and severity analysis. The integration of YOLO, CNN, and Grad-CAM improves detection accuracy, classification, performance, and model interpretability. The proposed system reduces manual effort and enables efficient large-scale monitoring. It can be effectively used in smart city applications for road maintenance. Future work can focus on real-time deployment and mobile-based implementation [9][10]. The system can be further enhanced by integrating real-time video processing and cloud-based deployment for large-scale monitoring. It also has potential applications in autonomous driving and smart city infrastructure.

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